

Predicting Diabetes Risk Using Health Data

(Technical Report)

1. Executive Summary

This project develops a machine learning model to predict the risk of diabetes using health-related survey data from the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset.

Diabetes is a major public health issue in the United States, and early identification of at-risk individuals can help prevent serious complications and reduce long-term healthcare costs.

The dataset includes over 250,000 observations and contains demographic, behavioral, and health-related variables such as BMI, age, blood pressure, cholesterol, physical activity, and general health. The original three-category diabetes variable was converted into a binary outcome to focus on identifying individuals who are at risk.

Three machine learning models were implemented and compared: Logistic Regression, Random Forest, and Gradient Boosting. The models were evaluated using Accuracy, F1 Score, and ROC AUC to ensure reliable performance, especially given the class imbalance in the dataset.

The results showed that Gradient Boosting performed the best, achieving a ROC AUC of 0.824, indicating strong predictive ability in distinguishing between individuals at risk and those not at risk. The most important predictors of diabetes risk were BMI, general health, age, high blood pressure, and cholesterol levels.

Based on these findings, it is recommended that healthcare providers prioritize early screening for individuals with higher BMI, poor general health, and cardiovascular risk factors. Targeted

prevention programs and lifestyle interventions can help reduce the progression of diabetes and improve patient outcomes.

Overall, this project demonstrates that machine learning models can effectively use survey-based health data to identify individuals at risk of diabetes. This approach can support healthcare organizations in improving early detection, optimizing resource allocation, and reducing healthcare costs through preventive care.

2. Introduction & Problem Definition

Diabetes is a serious and long-term health condition that affects millions of people in the United States and around the world. It occurs when the body is unable to properly regulate blood glucose levels, either due to insufficient insulin production or ineffective use of insulin. If not detected and managed early, diabetes can lead to severe complications, including cardiovascular disease, kidney failure, nerve damage, and vision loss. As a result, diabetes represents both a major clinical challenge and a significant public health concern.

One of the key challenges in managing diabetes is early detection. Many individuals, especially those with prediabetes, are unaware of their condition because symptoms may not appear in the early stages. Prediabetes is a critical phase where blood glucose levels are elevated but not yet at diabetic levels. Identifying individuals at this stage provides an opportunity for early intervention and prevention. Therefore, developing tools that can accurately identify individuals at risk is essential for improving health outcomes and reducing long-term healthcare costs.

In recent years, machine learning classification models have been widely used in healthcare to analyze large datasets and identify patterns that are not easily observable through traditional

methods. These models can use structured health data to predict disease risk and support clinical decision-making. In this project, machine learning techniques are applied to predict whether an individual is at risk of diabetes based on demographic, behavioral, and health-related indicators.

The objective of this project is to develop a predictive model that can classify individuals as either at risk or not at risk of diabetes using non-invasive health indicators. The dataset used in this analysis is derived from the Behavioral Risk Factor Surveillance System (BRFSS) 2015 survey and contains over 250,000 observations. It includes variables such as Body Mass Index (BMI), age category, physical activity, general health, high blood pressure, and cholesterol levels.

The original target variable, `Diabetes_012`, contains three categories: 0 (no diabetes), 1 (prediabetes), and 2 (diabetes). For the purpose of this analysis, the problem is simplified into a binary classification task. A new variable is created where 0 represents no diabetes and 1 represents prediabetes or diabetes. This transformation allows the model to focus on identifying individuals who are at risk.

The main research question for this project is:

Can machine learning models accurately predict whether an individual has or is at risk of diabetes based on health indicators?

To address this question, three supervised learning models are implemented and compared:

- Logistic Regression, as a baseline model
- Random Forest, to capture nonlinear relationships

- Gradient Boosting, to improve predictive performance

The dataset is divided into training and testing sets, where the models are trained on historical data and evaluated on unseen data to assess their generalization performance.

Model performance is evaluated using multiple classification metrics, including Accuracy, Precision, Recall, F1 Score, and ROC AUC. Among these, ROC AUC is the primary metric, as it measures the model's ability to distinguish between individuals at risk and those not at risk across different thresholds. The F1 Score is also important, as it balances precision and recall and ensures that at-risk individuals are correctly identified.

For this project, model success is defined by achieving a high ROC AUC score, indicating strong classification performance. In addition, a balanced F1 Score is used to ensure that the model effectively identifies at-risk individuals without excessive false predictions. A model achieving an ROC AUC above 0.80 is considered strong performance for this task.

In addition to predictive accuracy, this project also aims to identify the most important features associated with diabetes risk. Understanding the influence of variables such as BMI, age, and general health can provide valuable insights for healthcare providers and support targeted prevention strategies.

Overall, this project demonstrates how machine learning can be applied to a real-world healthcare problem. By predicting diabetes risk using non-invasive indicators, the results can support early detection, improve decision-making, and contribute to better health outcomes.

3. Data Overview

The dataset used in this project is the Diabetes Health Indicators dataset derived from the Behavioral Risk Factor Surveillance System (BRFSS) 2015 survey. It contains approximately 253,680 observations and 22 features, providing a large and representative sample for machine learning analysis. The data includes demographic, behavioral, and health-related variables collected through self-reported surveys.

The original target variable in the dataset is Diabetes_012, which contains three categories:

- 0 = No diabetes
- 1 = Prediabetes
- 2 = Diabetes

For this analysis, the target variable was transformed into a binary variable. A new variable, diabetes_binary, was created, where 0 represents no diabetes and 1 represents prediabetes or diabetes. This transformation simplifies the classification task and focuses on identifying individuals who are at risk.

Key predictor variables used in this analysis include BMI (Body Mass Index), age category, high blood pressure (HighBP), high cholesterol (HighChol), general health (GenHlth), physical activity (PhysActivity), and difficulty walking (DiffWalk). These features are non-invasive and commonly available, making them suitable for real-world applications such as early screening and prevention.

The dataset was evaluated for data quality issues, including missing values, inconsistencies, and duplicate records. The analysis showed that the dataset contains minimal missing values and no significant inconsistencies. Duplicate records were checked and removed where necessary.

Overall, the dataset is clean, well-structured, and suitable for predictive modeling.

Initial exploration of the data shows that the majority of individuals belong to the non-diabetic group, while a smaller portion are classified as at risk. This indicates the presence of class imbalance, which is an important consideration during model evaluation. In addition, BMI values are generally concentrated within a moderate range, with some higher values that may be associated with increased diabetes risk. These patterns are examined in more detail in the Exploratory Data Analysis (EDA) section.

In addition to data quality, ethical considerations were taken into account. The dataset is based on self-reported information, which may introduce bias or inaccuracies. There is also a risk that predictive models could reinforce existing health disparities if certain populations are underrepresented. Therefore, the results of this analysis should be used as a supportive tool rather than a replacement for professional medical judgment. Ensuring fairness and responsible use of the model is important when applying it in real-world healthcare settings.

In summary, the dataset provides a large, clean, and meaningful set of health-related features for predicting diabetes risk. Key characteristics such as class imbalance and important health indicators will influence the modeling process and are explored further in the next section.

4. Data Cleaning, Preprocessing, and Exploratory Data Analysis (EDA)

4.1 Data Cleaning

The dataset was first examined for data quality issues, including missing values, duplicate records, and inconsistencies. The analysis showed that the dataset contains little to no missing values, with less than 1% missing across all variables, which reduced the need for extensive data imputation. For numerical variables, median imputation was considered where necessary, although only a very small number of values required handling.

Duplicate records were identified and removed to ensure that each observation represents a unique individual. Data types were also verified to confirm that all variables were correctly formatted for analysis.

Outliers were examined using summary statistics and boxplots, particularly for continuous variables such as BMI. Some high BMI values (above 50) were observed; however, these were retained because they represent realistic cases and are important for identifying individuals at higher risk of diabetes.

Overall, the dataset was found to be clean, well-structured, and suitable for modeling.

4.2 Feature Engineering and Preprocessing

The original target variable, `Diabetes_012`, contained three categories: no diabetes (0), prediabetes (1), and diabetes (2). For this analysis, it was transformed into a binary variable (`diabetes_binary`), where 0 represents no diabetes and 1 represents prediabetes or diabetes.

After transformation, approximately 84% of individuals were classified as non-diabetic, while about 16% were classified as at risk. This confirms the presence of class imbalance.

Since most variables are already binary or numerically encoded, minimal feature engineering was required. All features were retained due to their relevance to diabetes risk.

Feature scaling was applied to Logistic Regression using standardization, while tree-based models (Random Forest and Gradient Boosting) were used without scaling.

4.3 Exploratory Data Analysis (EDA)

4.3.1 Target Distribution

The distribution of the target variable shows a clear class imbalance, with approximately 84% of individuals in the non-diabetic group and 16% in the at-risk group. This imbalance is important, as it may affect model performance and requires careful evaluation using metrics such as F1 Score and ROC AUC.

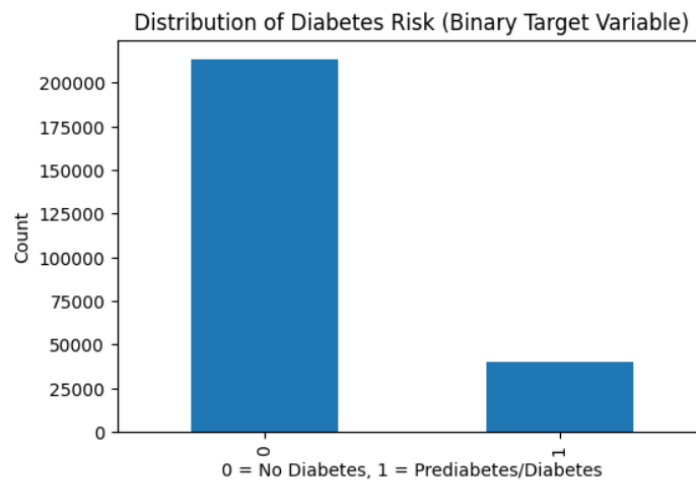


Figure 1. Distribution of diabetes risk (binary target).

4.3.2 BMI Distribution and Comparison

Figure 2 shows the distribution of BMI values. Most individuals have BMI values between 20 and 35, with the mean BMI approximately in the mid-20s. The distribution is right-skewed, with a smaller number of individuals having BMI values above 40.

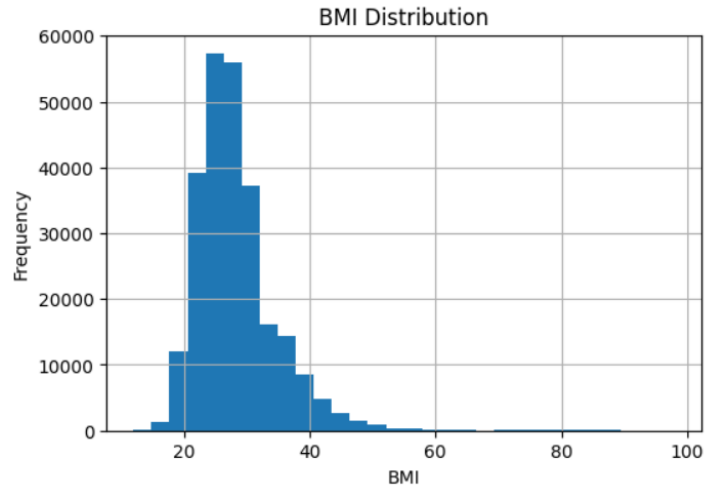


Figure 2. Distribution of Body Mass Index (BMI).

Figure 3 compares BMI values between groups. Individuals at risk have a noticeably higher median BMI (around 30) compared to the non-diabetic group (around 26). This indicates that higher BMI is strongly associated with increased diabetes risk.

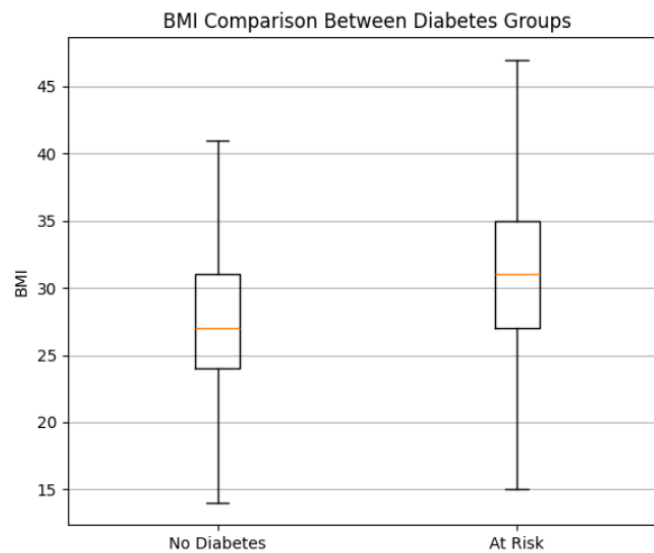


Figure 3. BMI comparison between diabetes groups.

4.3.3 Age and Diabetes Risk

Figure 4 shows that diabetes prevalence increases steadily across age categories. In the youngest age groups, the proportion at risk is below 5%, while in older groups it increases to over 20%.

This demonstrates a strong positive relationship between age and diabetes risk.

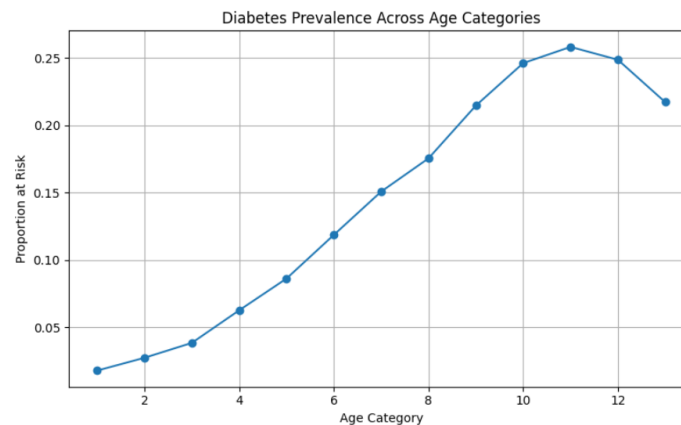


Figure 4. Diabetes prevalence across age categories.

4.3.4 General Health and Diabetes Risk

Figure 5 illustrates the relationship between general health and diabetes risk. Individuals reporting excellent health (category 1) have a low risk (below 5%), while those reporting poor health (category 5) show a much higher risk (above 30%). This indicates a strong relationship between general health status and diabetes.

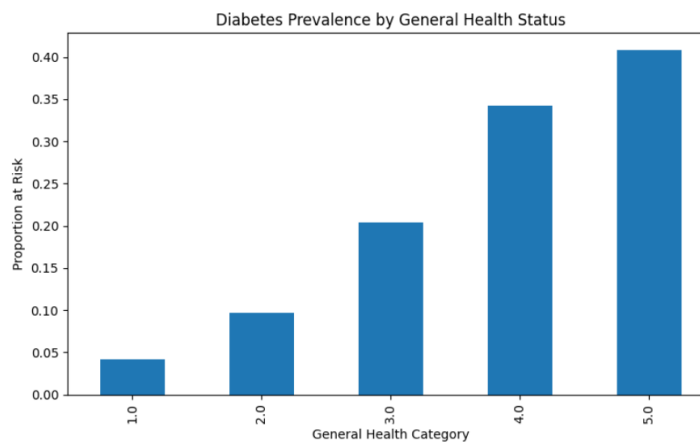


Figure 5. Diabetes prevalence by general health status.

4.3.5 Correlation Analysis

The correlation heatmap (Figure 6) shows that most variables have weak to moderate correlations, generally below 0.5, indicating low multicollinearity. Moderate positive relationships were observed between diabetes risk and variables such as BMI, general health, and physical health.

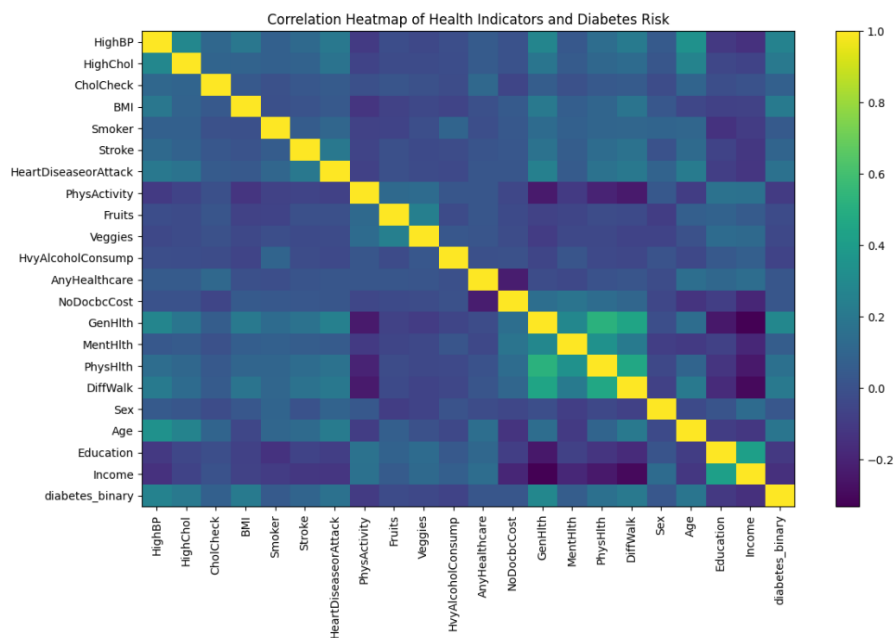


Figure 6. Correlation heatmap of key variables.

4.4 Statistical Insights

Statistical comparisons between groups show that individuals at risk have higher average BMI (approximately 30 vs 26), poorer general health scores, and more physical health limitations. These differences indicate meaningful effect sizes and confirm that these variables are strong predictors of diabetes risk.

4.5 Key Findings

The EDA results highlight several important patterns:

- Approximately 84% of individuals are non-diabetic, while 16% are at risk
- Individuals at risk have significantly higher BMI values
- Diabetes prevalence increases with age, exceeding 20% in older groups
- Poor general health is strongly associated with increased risk
- Physical health limitations contribute to higher risk
- No strong multicollinearity issues were detected

Overall, the dataset is clean, well-structured, and contains meaningful variables that are highly relevant for predicting diabetes risk.

5. Modeling / Analysis

This project is formulated as a supervised binary classification problem. The goal is to predict whether an individual has prediabetes or diabetes (1) versus no diabetes (0) using demographic, behavioral, and cardiovascular indicators. Because the outcome variable is labeled and binary, classification algorithms are appropriate for this task.

The objective of the modeling process is to compare linear and nonlinear models and evaluate whether more complex methods improve the prediction of diabetes risk.

5.1 Model Selection

Three models were implemented with increasing levels of complexity:

- Logistic Regression

Logistic Regression was used as a baseline linear classifier due to its simplicity and interpretability. It provides a useful benchmark and helps identify whether linear relationships are sufficient for this problem. Class weighting was applied to address class imbalance.

- Random Forest

Random Forest was selected as a bagging-based ensemble method that builds multiple decision trees and aggregates their predictions. This allows the model to capture nonlinear relationships and feature interactions while reducing variance.

- Gradient Boosting

Gradient Boosting was included as a boosting-based ensemble method that iteratively improves predictions by correcting previous errors. It is known for strong performance in structured tabular data and is often one of the most effective models for classification tasks.

Together, these models allow for comparison between a simple interpretable model and more advanced ensemble approaches.

5.2 Training Process and Data Splitting

The dataset was split into 80% training and 20% testing data using stratified sampling to preserve the class distribution. This ensures that both sets reflect the imbalance present in the data.

To improve the reliability of model evaluation, cross-validation was implemented using Stratified K-Fold validation. This method divides the training data into multiple subsets, allowing each model to be trained and validated across different data splits. The average

performance across folds provides a more stable estimate of model performance and reduces the risk of overfitting.

5.3 Feature Scaling

For Logistic Regression, features were standardized using z-score scaling to ensure that all variables are on a similar scale. This improves model convergence and stability.

For Random Forest and Gradient Boosting, no scaling was applied, as tree-based models are not sensitive to feature magnitude.

5.4 Evaluation Metrics

Model performance was evaluated using multiple metrics:

- Accuracy – measures overall correctness
- F1 Score – balances precision and recall
- ROC AUC – primary evaluation metric

ROC AUC was selected as the main success criterion because it evaluates how well the model distinguishes between individuals at risk and those not at risk across different classification thresholds.

Figure 7 shows the ROC curve comparison for all models. Gradient Boosting achieved the highest ROC AUC (0.824), followed by Logistic Regression (0.817) and Random Forest (0.795), indicating that Gradient Boosting provides the strongest classification performance.

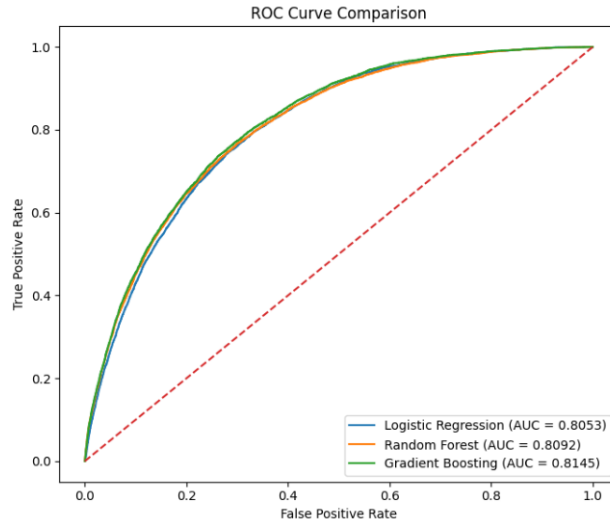


Figure 7. ROC curve comparison of Logistic Regression, Random Forest, and Gradient Boosting models.

5.5 Hyperparameter Tuning

Hyperparameter tuning was performed to optimize model performance.

For Logistic Regression, the number of iterations and class weighting were adjusted to improve convergence and handle class imbalance.

For Random Forest, parameters such as the number of trees (`n_estimators`), maximum depth, and minimum samples per split were tuned to control overfitting and improve model stability.

For Gradient Boosting, parameters including learning rate, number of estimators, and tree depth were tuned. Lower learning rates combined with a higher number of estimators improved performance while maintaining model stability.

5.6 Model Comparison and Interpretation

The performance results show that Gradient Boosting achieved the best overall performance based on ROC AUC, followed by Logistic Regression and Random Forest.

Although Logistic Regression achieved a higher F1 Score, Gradient Boosting provided better overall class separation. This suggests that nonlinear relationships between features improve predictive performance.

Figure 8 presents the confusion matrix for the best-performing model (Gradient Boosting). The model correctly identifies a large portion of non-diabetic individuals, while performance on the at-risk group is lower, reflecting the impact of class imbalance.

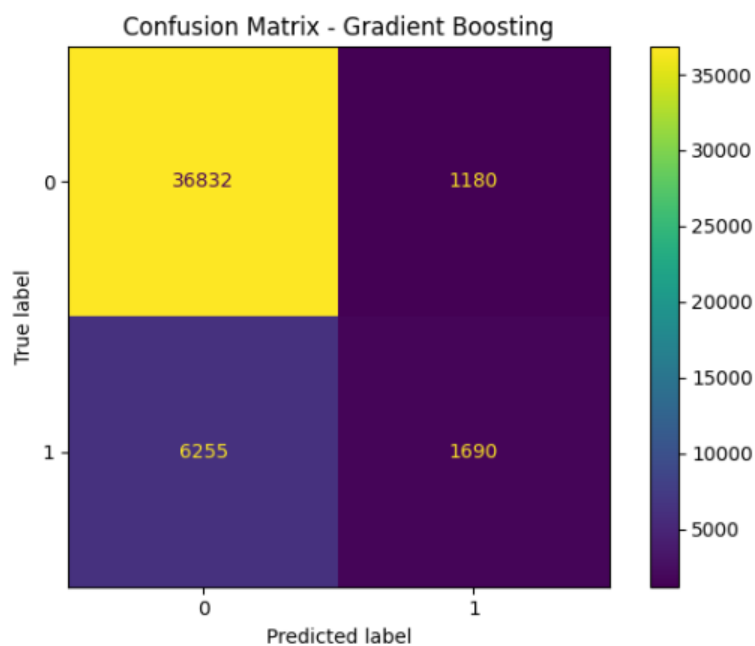


Figure 8. Confusion matrix for the Gradient Boosting model.

5.7 Feature Importance

To understand which variables, drive predictions, feature importance was analyzed for the Gradient Boosting model.

Figure 9 shows the top features contributing to diabetes risk prediction. BMI, general health, age, high blood pressure, and cholesterol were identified as the most important variables.

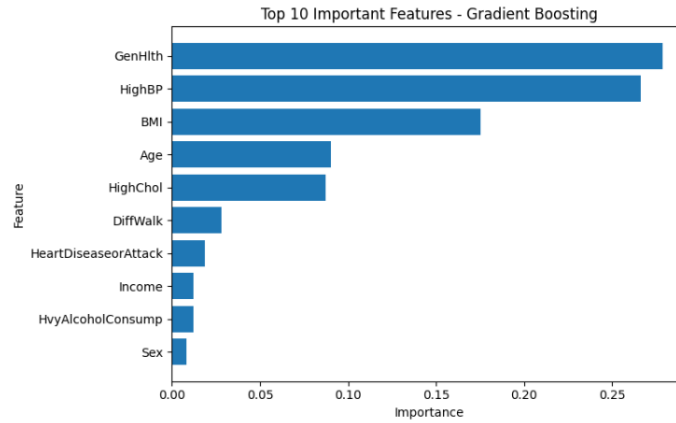


Figure 9. Top feature importance from the Gradient Boosting model.

These results confirm that both lifestyle factors and existing health conditions play a major role in determining diabetes risk.

5.8 Summary of Modeling Approach

Overall, the modeling approach demonstrates that ensemble methods outperform the linear baseline for this task. Gradient Boosting achieved the strongest predictive performance and provides a reliable model for identifying individuals at risk of diabetes.

The inclusion of cross-validation, hyperparameter tuning, and multiple evaluation metrics ensures that the results are robust and generalizable.

The next section presents a detailed analysis of model performance and interprets the results in the context of healthcare decision-making.

6. Results

This section presents the performance of the machine learning models and evaluates their effectiveness in predicting diabetes risk. The models were compared using Accuracy, F1 Score,

ROC AUC, and cross-validation performance, with ROC AUC serving as the primary evaluation metric.

6.1 Model Performance Comparison

Table X presents the performance comparison of Logistic Regression, Random Forest, and Gradient Boosting.

Model	Accuracy	F1 Score	ROC AUC	CV ROC AUC (Mean)	CV ROC AUC (Std)
Gradient Boosting	0.838	0.313	0.815	0.809	0.0020
Random Forest	0.836	0.208	0.809	0.805	0.0019
Logistic Regression	0.720	0.480	0.805	0.801	0.0023

Gradient Boosting achieved the best overall performance, with a ROC AUC of 0.8145, followed by Random Forest (0.8092) and Logistic Regression (0.8053). This indicates that Gradient Boosting has the strongest ability to distinguish between individuals at risk and those not at risk.

In terms of accuracy, Gradient Boosting also performed the best (0.838), followed closely by Random Forest (0.836), while Logistic Regression showed lower accuracy (0.720).

However, Logistic Regression achieved the highest F1 Score (0.480), compared to Gradient Boosting (0.313) and Random Forest (0.208). This indicates that Logistic Regression performs better in balancing precision and recall for the minority class.

Cross-validation results further confirm model stability. Gradient Boosting achieved a mean CV ROC AUC of 0.8094 with very low variance (± 0.0020), indicating consistent performance across different data splits.

Overall, Gradient Boosting was selected as the best-performing model due to its superior ROC AUC and stable cross-validation performance.

6.2 Interpretation of Model Performance

The results highlight an important trade-off between overall performance and minority class detection.

While Gradient Boosting provides the highest overall accuracy and ROC AUC, it performs less effectively in identifying at-risk individuals, as reflected in its lower recall for the minority class. Logistic Regression, on the other hand, achieves a higher F1 Score, indicating better balance between precision and recall. This suggests that simpler models may still be valuable when the goal is to identify at-risk individuals.

These findings demonstrate that model selection depends on the specific objective—whether maximizing overall accuracy or improving detection of high-risk cases.

6.3 Feature Importance and Key Drivers

Feature importance analysis from the Gradient Boosting model shows that the most influential predictors are:

- General Health (GenHlth)
- High Blood Pressure (HighBP)
- BMI
- Age
- High Cholesterol (HighChol)

These variables have the strongest impact on predicting diabetes risk.

The importance of these features aligns with real-world medical knowledge. Poor general health, obesity, and cardiovascular conditions are well-known risk factors for diabetes.

6.4 Connection to Data Patterns

The modeling results are consistent with patterns observed during exploratory data analysis.

Diabetes risk increases significantly with age and is strongly associated with higher BMI and poorer general health. These relationships help explain why these variables are highly important in the model.

This consistency between EDA and modeling strengthens confidence in the results.

6.5 Comparative Analysis of Models

The comparison of models reveals that ensemble methods (Random Forest and Gradient Boosting) outperform Logistic Regression in terms of ROC AUC and overall accuracy.

This suggests that nonlinear relationships and interactions between variables are important for predicting diabetes risk.

However, Logistic Regression performs better in terms of F1 Score, indicating stronger performance in identifying the minority class.

Overall, Gradient Boosting provides the best balance between predictive performance and generalization, making it the preferred model.

6.6 Limitations

Several limitations should be considered.

The dataset is based on self-reported survey data, which may introduce bias or inaccuracies.

Additionally, the dataset does not include clinical measurements such as blood glucose levels, which could improve model performance.

Class imbalance also affects the model's ability to detect at-risk individuals, as seen in the lower recall for this group.

6.7 Business Implications

The results demonstrate that machine learning can be used to identify individuals at risk of diabetes using simple, non-invasive health indicators.

Healthcare providers can use these insights to prioritize screening for individuals with high BMI, poor general health, and cardiovascular risk factors.

This can support early intervention, improve patient outcomes, and reduce long-term healthcare costs.

6.8 Summary of Results

Overall, Gradient Boosting achieved the best performance, with an ROC AUC of 0.8145 and strong accuracy.

The analysis confirms that BMI, general health, age, and cardiovascular indicators are key predictors of diabetes risk.

While limitations exist, the models provide valuable insights and demonstrate the potential of data-driven approaches in healthcare.

7. Recommendations

Based on the results of this analysis, several practical recommendations can be made for healthcare providers and public health organizations.

7.1 Targeted Screening and Early Detection

Healthcare providers should prioritize early screening for individuals who show high-risk characteristics identified by the model. In particular, individuals with high BMI, poor general health, high blood pressure, and older age should be considered high priority for screening.

Using these indicators, healthcare systems can identify at-risk patients earlier and recommend preventive actions before the condition progresses to diabetes.

7.2 Preventive Health Programs

Public health programs should focus on reducing modifiable risk factors. The results show that lifestyle-related variables such as BMI and physical health play a major role in diabetes risk.

Programs should promote:

- Increased physical activity
- Healthy eating habits
- Weight management
- Regular health check-ups

These interventions can significantly reduce the risk of developing diabetes and improve overall population health.

7.3 Use of Predictive Models in Healthcare Systems

The Gradient Boosting model developed in this project can be used as a decision-support tool in healthcare settings.

For example, clinics and hospitals can integrate this model into electronic health record systems to automatically flag individuals who are at higher risk. This would allow healthcare providers to focus attention on patients who need early intervention.

It is important to note that the model should support—not replace—clinical decision-making.

7.4 Implementation Considerations

Several factors should be considered when implementing these recommendations.

First, data quality is important. Since the dataset is based on self-reported information, healthcare providers should ensure that collected data is accurate and reliable.

Second, ethical considerations must be addressed. The use of predictive models should avoid bias and ensure fairness across different populations.

Third, model performance should be monitored regularly. As new data becomes available, the model should be updated and re-evaluated to maintain accuracy.

7.5 Future Work and Improvements

Future work can improve the model in several ways.

Additional datasets that include clinical measurements, such as blood glucose levels, could significantly improve predictive accuracy.

More advanced modeling techniques, such as deep learning or ensemble stacking methods, could also be explored.

In addition, addressing class imbalance using techniques such as resampling or cost-sensitive learning may improve detection of at-risk individuals.

Finally, testing the model on different populations would help evaluate its generalizability and real-world applicability.

7.6 Summary of Recommendations

Overall, healthcare providers should focus on early identification of high-risk individuals, promote preventive health programs, and consider integrating machine learning models into their systems.

By taking these actions, organizations can improve early detection, reduce the burden of diabetes, and support better long-term health outcomes.

8. Overall Narrative, Communication, and Conclusion

This project aimed to predict diabetes risk using simple health indicators such as BMI, age, and general health. The analysis followed a clear process, starting with data cleaning and exploration, then building and comparing machine learning models.

The results showed clear patterns in the data. People with higher BMI, poorer general health, and older age were more likely to be at risk. These findings were consistent in both the data analysis and the model results.

Among the models tested, Gradient Boosting performed the best, achieving the highest ROC AUC (0.8145). This shows that more advanced models can better capture complex relationships in the data. However, Logistic Regression performed better in detecting at-risk individuals, showing that different models have different strengths.

The results are useful in real-world settings because they rely on simple, non-invasive data. Healthcare providers can use these findings to identify high-risk individuals early and focus on prevention.

There are some limitations. The data is self-reported, which may not always be accurate, and class imbalance makes it harder to detect at-risk individuals.

In conclusion, this project shows that diabetes risk can be predicted using basic health data. The results support the use of machine learning as a tool to improve early detection and help guide better healthcare decisions.

9. AI Appendix

ChatGPT was used to help improve the writing and organization of this report. It was mainly used to make the language clearer, structure the sections properly, and format tables and figures.

AI was not used to perform any data analysis, build models, or generate results. All coding, data processing, and model evaluation were completed by the author.

Understanding Diabetes Risk Using Health Data

(Non-Technical Report)

1. Executive Summary

This project aims to identify people who may be at risk of diabetes using simple health information such as body weight (BMI), age, and general health. Diabetes is a serious health condition, and early detection can help prevent complications and reduce healthcare costs.

The analysis used a large dataset of over 250,000 survey responses that include information about people's health, lifestyle, and background. Machine learning models were used to find patterns in the data and predict whether someone is at risk.

Among the models tested, Gradient Boosting performed the best, showing strong ability to correctly identify individuals at risk of diabetes. The most important factors were BMI, general health, age, blood pressure, and cholesterol.

The results suggest that healthcare providers can use this type of data to identify high-risk individuals earlier. This can help focus screening efforts, improve prevention programs, and support healthier lifestyle choices.

Overall, this project shows that simple health information can be used effectively to detect diabetes risk and support better healthcare decisions.

2. Introduction

Diabetes is a serious health condition that affects many people in the United States and around the world. It happens when the body cannot properly control blood sugar levels. If diabetes is not

detected early, it can lead to serious problems such as heart disease, kidney damage, and vision loss. Because of this, early detection is very important.

One of the main challenges is that many people do not know they have diabetes or prediabetes.

Prediabetes is an early stage where blood sugar is higher than normal but not yet at a dangerous level. Since symptoms are often not clear, many people are unaware of their condition.

Identifying these individuals early can help prevent the disease from getting worse.

The goal of this project is to find a way to identify people who are at risk of diabetes using simple health information. This includes factors such as body weight (BMI), age, general health, blood pressure, and lifestyle habits like physical activity. These are all things that can be collected easily without medical tests.

The data used in this project comes from a large health survey conducted in 2015, which includes information from over 250,000 people. By analyzing this data, patterns can be found that help predict whether someone is likely to be at risk.

To do this, different computer models were used to learn from the data and make predictions.

These models look for patterns and relationships between health factors and diabetes risk. The results showed that the models were able to identify at-risk individuals with good accuracy.

The analysis also found that some factors are more important than others. For example, people with higher body weight, poorer general health, and older age are more likely to be at risk of diabetes.

Based on these findings, it is recommended that healthcare providers focus on early screening for individuals with these risk factors. Encouraging healthier lifestyles, such as better diet and increased physical activity, can also help reduce the risk.

Overall, this project shows that simple health information can be used to identify people at risk of diabetes. This can help improve early detection, support prevention efforts, and lead to better health outcomes.

3. Data overview:

The data used in this project comes from a national health survey in the United States. It includes over 250,000 people and contains information about their health, lifestyle, and background.

The data was collected through a large national survey, where individuals reported information about their health and lifestyle. This type of data provides a broad view of the population but may include some inaccuracies due to self-reporting.

Some of the most important factors in the data include:

- Body Mass Index (BMI)
- Age
- Blood pressure
- Cholesterol
- General health
- Physical activity

The data is clean and mostly complete, with very few missing values.

One important observation is that most people in the dataset are not diabetic, while a smaller group is at risk. This imbalance is important and affects how the models are evaluated.

4. Data Cleaning, Preprocessing, Exploratory Data Analysis (EDA)

Before building any models, the data was checked to make sure it was clean and reliable.

First, the dataset was examined for missing values. Only a very small number of missing values were found, so no major changes were needed. This means the data was mostly complete and ready to use.

Next, duplicate records were checked and removed to make sure each row represents a unique individual. The data was also reviewed for any unusual or incorrect values.

Some very high BMI values were observed, but these were kept because they are realistic and important for identifying people at higher risk.

After cleaning, the data was explored to understand important patterns.

Several clear trends were found:

- People with higher BMI are more likely to be at risk of diabetes
- Diabetes risk increases with age
- People who report poor general health have much higher risk
- Individuals with physical health problems are also more likely to be at risk

These patterns are important because they help explain which factors are most related to diabetes risk.

Overall, the dataset was clean and provided useful information. The patterns found in this step helped guide the model building and support the final results.

5. Modeling/Analysis

To answer the main question of this project, different prediction methods were tested to see which one works best for identifying people at risk of diabetes.

The data was divided into two parts. One part was used to learn patterns, and the other part was used to

test how well those patterns work on new data. This helps make sure the results are reliable.

Several different approaches were compared. Some were simple and easy to understand, while others were more advanced and able to capture more complex patterns in the data.

Each approach was evaluated based on how well it could correctly identify people who are at risk and those who are not.

The results showed that more advanced methods performed better overall because they were able to capture more detailed relationships between health factors.

However, simpler approaches were still useful because they were better at identifying some at-risk individuals.

The reason different approaches perform differently is because some methods are better at finding simple patterns, while others can capture more complex relationships in the data. This explains why the more advanced methods performed better overall in this project.

Overall, testing multiple approaches helped ensure that the final results are accurate and reliable, and it allowed for a better understanding of how different methods perform.

6. Results

The goal of this project was to predict whether a person is at risk of diabetes using simple health information. Several models were tested, and their performance was compared.

The results showed that the Gradient Boosting model performed the best overall. This model was better at identifying patterns in the data and separating people who are at risk from those who are not. It achieved the highest overall performance compared to the other models.

However, each model had its own strengths. The Logistic Regression model performed better at identifying people who are at risk, meaning it was more balanced when detecting the smaller group. This is important because identifying at-risk individuals is often more valuable than simply achieving high overall accuracy.

The results also showed that the models performed very well in identifying people who are not at risk.

However, they had more difficulty correctly identifying people who are at risk. This is mainly because the dataset contains many more non-diabetic individuals than at-risk individuals.

Several important factors were identified as strong predictors of diabetes risk. These include:

- Body Mass Index (BMI)
- General health status
- Age
- High blood pressure
- High cholesterol

People with higher BMI values were more likely to be at risk. Similarly, individuals who reported poor general health had a much higher chance of being classified as at risk. Age also played a significant role, with older individuals showing higher risk levels.

These results are consistent with real-world medical knowledge, which increases confidence in the model's findings.

These results are important because they show that simple health information can be used to identify people at risk before serious symptoms appear. This means healthcare providers can take action earlier, such as recommending lifestyle changes or further testing. Even though the model is not perfect, it provides a useful starting point for improving early detection and prevention.

Overall, the models were able to successfully identify key patterns in the data and provide useful predictions. While not perfect, the results show that simple health information can be used to estimate diabetes risk with reasonable accuracy.

7. Recommendations

Based on the results of this analysis, several practical recommendations can be made for healthcare providers and public health organizations.

First, healthcare providers should focus on early screening for individuals who show high-risk characteristics. People with high BMI, poor general health, high blood pressure, and older age should be

prioritized for further testing and monitoring.

Early detection is important because it allows for intervention before the condition becomes more serious.

By identifying at-risk individuals earlier, healthcare providers can help prevent the development of diabetes.

Second, public health programs should promote healthier lifestyles. Since many of the key risk factors are related to lifestyle, such as weight and physical health, programs should focus on:

- Encouraging regular physical activity
- Promoting healthy eating habits
- Supporting weight management
- Increasing awareness about diabetes risk

These programs can help reduce the number of people who develop diabetes in the future.

Third, healthcare systems can use predictive models like the one developed in this project as a support tool. For example, hospitals or clinics could use these models to flag patients who may be at higher risk.

This would help doctors focus their attention on patients who need it most.

However, it is important that these models are used carefully. They should not replace doctors, but rather support decision-making.

There are also some important considerations for implementation. The data used in this project is based on self-reported information, which may not always be accurate. Healthcare providers should ensure that reliable data is used when applying similar models.

In addition, fairness and ethics should be considered. Models should be tested on different populations to make sure they work equally well for everyone.

Finally, future improvements can make the model even better. Adding clinical data, such as blood sugar levels, could improve accuracy. New modeling techniques could also be tested to improve performance, especially for identifying at-risk individuals.

In summary, healthcare providers should focus on early detection, promote healthy lifestyles, and

consider using predictive models as support tools. These actions can help improve health outcomes and reduce the impact of diabetes.

8. Conclusion

This project set out to answer a simple but important question: can we use everyday health information to identify people at risk of diabetes?

The results show that we can. By using data such as body weight, age, and general health, it is possible to find clear patterns that help identify individuals who may be at higher risk.

One of the most important findings is that people with higher BMI, poorer general health, and older age are more likely to be at risk. These are factors that are easy to measure and already available in most healthcare settings, which makes the approach practical and useful.

While no model is perfect, the analysis shows that data can be a powerful tool to support early detection. Instead of waiting until symptoms become serious, healthcare providers can take action earlier and focus on prevention.

This project highlights how data-driven methods can support better health decisions. By identifying high-risk individuals early, it becomes possible to improve outcomes, reduce complications, and lower healthcare costs.

In the end, the key message is simple: small pieces of information can make a big difference when used in the right way. Using data wisely can help people live healthier lives and reduce the impact of diseases like diabetes.

9. AI Appendix

ChatGPT was used to help improve the writing and organization of this report. It was mainly used to simplify the language and structure.

AI was not used for analysis or model building. All data work and results were completed by the author.